

Collins Kimathi Mwiti

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Nairobi

[Medium](#)

PROFILE SUMMARY

An analytical and results-driven professional with expertise in soil science, crop management, and sustainable Agriculture field practices. Skilled at providing data-based technical guidance to farmers and delivering effective training. Experienced in using digital tools like GIS and remote sensing to improve land use and crop yields. Focused on helping farmers and communities make better decisions through clear, evidence-based data.

EDUCATION

Land Resource Planning and Management

Jomo Kenyatta University of Agriculture and Technology • Kiambu, Kenya

04/2021 Second class Upper division Hons

SKILLS

- Crop Nutrition Management: data-driven guidance on Crop Yield Optimization, Soil Health Analysis (N, P, K, pH), and efficient Resource Utilization.
- Crop Protection: Pest & Disease Management; Identification, Diagnosis, and Control Measures. Integrated Pest Management (IPM)
- Training and Capacity Building: Designing technical training for extension agents and smallholder farmer groups.
- Data Analytics: Data Collection, Cleaning, Analysis, and Validation (Kobo Collect, Google forms, Excel, SQL, Python)
- Remote Sensing: Vegetation health monitoring (NDVI), land cover mapping, and spatial analysis(Google earth Engine, QGIS, QSWAT)
- Geospatial Database Management: Design of geospatial databases and field data collection frameworks.
- Impact Visualization: Creating interactive stakeholder dashboards and Data storytelling (Tableau, PowerBI)
- Programming: Python (Pandas, Numpy, Matplotlib), Django, Reactjs
- Git/GitHub (version control for code/documentation)
- Reporting: Automated Reporting Processes, Technical Writing and reporting, Presentation Skills, proposal Development, Technical Documentation Development
- Project Coordination: Technical supervision of field operations, resource mapping, and inventory management.

WORK EXPERIENCE

Agronomist

Ujuzi Kilimo Solutions Nairobi, Kenya

11/2024 - Present

- Contributed to the design and expansion of the Farm Suite App, integrating a robust fertilizer recommendation engine and agronomic best practices library - Through this initiative, I directly impacted over 10,000 farmer interactions and scaled our digital advisory reach.
- Led data-driven soil fertility recommendations by analyzing thousands of farmer soil samples (N, P, K, pH, EC), improving crop-specific advisory services across Kenya and increasing farmer satisfaction with yield outcomes.
- I led training sessions for field agents and farmers, enhancing knowledge in soil health management, crop nutrition, and climate-smart agronomic practices, resulting in improved adoption of recommendations on the ground.
- I Collaborated with the hardware and software teams to calibrate and validate our spectral-based soil pal: a soil testing sensor. This helped us ensure accurate measurement of key parameters (N, P, K, pH, EC) against laboratory standards.
- Developed an interactive NDVI Explorer using Google Earth Engine for monitoring vegetation health and tracking seasonal crop performance across selected counties, supporting data-driven decision-making by the field teams and agronomists.
- I Built a predictive crop disease classification model using Python, - I integrated image and field data to enable early detection and response strategies; developed accompanying guidelines for identification, prevention, and management of major pests and diseases.

Project Lead (Impact Assessment) – ENI Partnership Meru, Kenya

07/2024 – 10/2024

- Sector-Specific Data Collection: Led a 3-month impact assessment for the Castor Oil farming project, focusing on smallholder farmer productivity and project socio-economic outcomes.
- Quantitative & Qualitative Surveys: Led the end-to-end data collection workflow, utilizing structured questionnaires to track agricultural outputs and farmer livelihoods.
- Field Coordination & Quality Assurance: Supervised field teams to ensure all survey responses were captured accurately in real-time, maintaining high data integrity for the ENI partnership team.
- Data Cleaning & Validation: Managed the verification of field data, identifying trends in crop performance and farmer participation; this was used in the final project evaluation reports.

Crop Production Manager

LumiGold LLC • Kajiado, Kenya

11/2022 - 06/2024

- Export Quality Control: Managed full production cycles for onions, cabbages, and French beans, ensuring 100% compliance with international buyer standards for export.
- Crop Nutrition & Health: Designed fertilization and pest control schedules specifically for the soils and climate of Kajiado, focusing on plant health and high-quality harvests.
- Resource Efficiency: Tracked water and fertilizer use to identify waste, successfully reducing costs while maintaining optimal soil moisture for the crops.
- Operations & Finance: Handled the farm's budget, managed procurement of seeds and inputs, and supervised a diverse field crew to meet strict planting and harvest deadlines.
- Market Coordination: Managed the relationship with exporters and local markets, ensuring harvest cycles were timed perfectly to meet demand and maximize profitability.

Technical Agriculture Fellow (AICAT Program)

Boaz Pepper • Paran Arava Israel

11/2021 - 08/2022

- Precision Desert Farming: Gained intensive field experience in the Arava region, mastering advanced irrigation and crop management techniques in a hyper-arid environment.
- Cooperative Farm Management: Studied and worked under the Moshav and Kibbutz systems, gaining practical insights into organized community farming.
- Applied Research: Conducted academic and field research on sustainable desert agriculture, focusing on how modern methods impact local ecosystems and resource conservation.
- Data-Led Optimization: Collected and analyzed field data on soil moisture, weather patterns, and crop performance to adjust irrigation and fertilization for maximum efficiency.
- Integrated Pest Management (IPM): Implemented standard pest and disease control measures required for high-value vegetable production in greenhouses and open fields.

Project Data Analyst

E4Impact • Isiolo, Kenya

02/2021 - 05/2021

- Geospatial Resource Mapping: Executed a large-scale field survey to map livestock distribution and water resources across Garbatulla, Merti, and Isiolo sub-counties using GPS technology.
- Data Visualization & GIS: Processed data for over 5,000 livestock units to create thematic maps and trend analyses, providing a visual framework for regional resource management.
- Decision Support for Government: Translated complex spatial data into actionable insights for local government and community leaders to prioritize development interventions and grazing management.
- Field Data Integrity: Managed the collection, cleaning, and validation of field data in remote environments, ensuring high accuracy for sustainable development planning.

Research Analyst

JKUATES • Samburu, Kenya

08/2020 - 07/2021

- Methodology Design: Collaborated with environmental scientists to design data collection methodologies, focusing on environmental impacts and conservation strategies in Samburu and Isiolo counties.
- Geospatial Data Analysis: Analyzed and interpreted geospatial data to assess water availability, Distribution, water Quality, and conservation strategies in Samburu, Kenya.
- Data Visualization: Created interactive dashboards to visualize water distribution patterns, quality metrics, facilitating informed decision-making for sustainable resource management.
- Decision Support for the county Government and partners: Translated complex spatial data into actionable insights for local government and community leaders to prioritize development interventions and water management.

Referees

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